

BUSINESS ANALYTICS

Level 2 Task 1: Data Visualisation with Power BI

Codveda Technology Internship Programme

June 2026

1. Introduction

Data visualisation transforms raw data into actionable insights by presenting complex patterns and relationships in a visually intuitive format. Power BI, Microsoft's business intelligence platform, enables analysts to connect diverse datasets, build interactive dashboards, and communicate findings to both technical and non-technical audiences through a single reporting environment.

This task involved building two fully interactive Power BI dashboards using real-world datasets: a Telecom Customer Churn dashboard and an Iris Species Classification dashboard. Both were constructed within a single .pbix file, demonstrating the ability to manage multiple data sources and report pages within one Power BI project.

2. Tools and Environment

Component	Detail
Platform	Microsoft Power BI Desktop
File Format	.pbix (Power BI project file)
Datasets	churn-bigml-80.csv (Telecom Churn), iris.csv (Iris Species)
Visualisation Types	Bar Charts, Pie Charts, Line Charts, Scatter Plots, KPI Cards, Slicers
Interactivity	Cross-filtering, slicers, page navigation, tooltips

3. Dashboard 1 — Telecom Customer Churn

3.1 Objective

To build an interactive dashboard that visualises customer churn patterns in a telecom dataset, enabling business stakeholders to identify at-risk customer segments based on usage behaviour and account characteristics.

3.2 Dataset Overview

The telecom churn dataset (churn-bigml-80.csv) contains 2,666 customer records across 20 features covering account details, usage statistics, and service plan subscriptions. The target variable is Churn (True/False), indicating whether a customer left the service.

Feature Category	Key Variables	Purpose
Account Details	State, Account Length, Area Code	Customer segmentation by geography and tenure
Service Plans	International Plan, Voice Mail Plan	Subscription-based churn drivers
Usage Statistics	Day/Evening/Night Charge, Intl Charge	Call behaviour and cost patterns
Service Interactions	Customer Service Calls	Proxy for customer dissatisfaction
Target Variable	Churn (True/False)	Binary outcome for churn classification

3.3 Dashboard Design and Visuals

The Churn Dashboard was structured to provide a complete overview of churn distribution and its key drivers across four visual panels:

- **KPI Cards** — Displayed total customers, total churned customers, and the overall churn rate as headline metrics at the top of the report page, giving stakeholders instant access to the most critical figures.
- **Churn by State (Bar Chart)** — Identified the geographic distribution of churned customers, highlighting states with disproportionately high churn rates as potential areas for targeted retention campaigns.
- **Churn by International Plan (Pie Chart)** — Revealed that customers with an international plan churned at significantly higher rates relative to those without, suggesting pricing or service quality issues specific to international callers.
- **Churn by Customer Service Calls (Line/Bar Chart)** — Showed a clear positive relationship between the number of customer service calls and churn probability, with customers making 4 or more service calls being far more likely to churn — a critical operational insight for retention teams.
- **Interactive Slicers** — State and International Plan slicers were added to allow dynamic filtering across all visuals simultaneously, enabling stakeholders to drill into specific segments without switching between report pages.

3.4 Key Insights

Insight	Finding	Business Implication
Overall Churn Rate	Approximately 14–15% of customers churned	Significant revenue risk; retention strategy needed
International Plan	Higher churn concentration among international plan holders	Review international plan pricing and service quality
Customer Service Calls	Churn risk escalates sharply after 3+ service calls	Flag high-contact customers for proactive outreach

Insight	Finding	Business Implication
Geographic Variance	Certain states showed above-average churn concentration	Deploy state-specific retention offers

4. Dashboard 2 — Iris Species Classification

4.1 Objective

To build an exploratory visualisation dashboard for the Iris dataset that communicates how petal and sepal measurements differ across the three Iris species — Setosa, Versicolor, and Virginica highlights the separability of species based on physical dimensions.

4.2 Dataset Overview

The Iris dataset contains 150 records across 5 columns: four numerical measurement features (sepal length, sepal width, petal length, and petal width, all in centimetres) and one categorical target column (species). Each of the three species is represented by exactly 50 records, making it a perfectly balanced dataset.

Feature	Type	Description
Sepal Length (cm)	Numerical	Length of the flower sepal
Sepal Width (cm)	Numerical	Width of the flower sepal
Petal Length (cm)	Numerical	Length of the flower petal
Petal Width (cm)	Numerical	Width of the flower petal
Species	Categorical	Target: Setosa, Versicolor, or Virginica

4.3 Dashboard Design and Visuals

The Iris Dashboard focused on comparative species analysis and feature distribution across four visualisation panels:

- Average Measurements by Species (Clustered Bar Chart) — Displayed the average petal length, petal width, sepal length, and sepal width for each species side by side. The chart immediately revealed that Setosa has distinctly smaller petals, while Virginica has the largest average petal dimensions visually reinforcing the separability observed in machine learning analysis.
- Petal Length vs Petal Width Scatter Plot — The most insightful visual, plotting each of the 150 data points with species as the colour dimension. Setosa formed a completely isolated cluster in the bottom-left corner, while Versicolor and Virginica clusters partially overlapped directly corroborating the KMeans clustering boundary behaviour observed in the machine learning tasks.

- Species Count Cards / Donut Chart — Confirmed the perfectly balanced dataset with 50 records per species, giving viewers confidence that no class imbalance would skew visual comparisons.
- Species Slicer — A species filter slicer allowed stakeholders to isolate any one species across all report visuals simultaneously, enabling focused analysis of individual species measurement profiles.

4.4 Key Insights

Insight	Finding	Analytical Implication
Setosa Separability	Setosa petals are markedly smaller than the other two species	Setosa is completely linearly separable — consistent with ML clustering results
Petal Dimensions as Discriminators	Petal length and width show stronger between-species separation than sepal dimensions	Petal features are the most important predictors for species classification
Versicolor/Virginica Overlap	The two species share overlapping petal measurement ranges at their boundaries	Explains the partial cluster mixing observed in the KMeans analysis
Dataset Balance	50 records per species	No class imbalance; unbiased visual comparisons across species

5. Consolidated Summary

Both dashboards were delivered within a single .pbix file, demonstrating efficient multi-dataset management within one Power BI project. The table below summarises the deliverables across both dashboard components:

Component	Churn Dashboard	Iris Dashboard
Dataset	churn-bigml-80.csv (2,666 rows, 20 cols)	iris.csv (150 rows, 5 cols)
Report Pages	1 dedicated page	1 dedicated page
Visual Types Used	KPI Cards, Bar, Pie, Line, Slicers	Bar, Scatter Plot, Donut, Slicers
Interactivity	State & Plan slicers with cross-filtering	Species slicer with cross-filtering
Primary Insight	Customer service calls drive churn risk	Petal dimensions separate Iris species

6. Conclusion

This task successfully demonstrated the application of Power BI to build interactive, business-ready dashboards from two distinct datasets. The Churn Dashboard delivered operational insights directly actionable for a telecom retention team, while the Iris Dashboard provided an exploratory

visual analysis that aligned closely with findings from the machine learning components of this internship programme.

Together, both dashboards reinforce the role of data visualisation as a bridge between raw analytical outputs and business decision-making translating statistical patterns into intuitive visual narratives accessible to non-technical stakeholders.

